

30264 - Public Finance

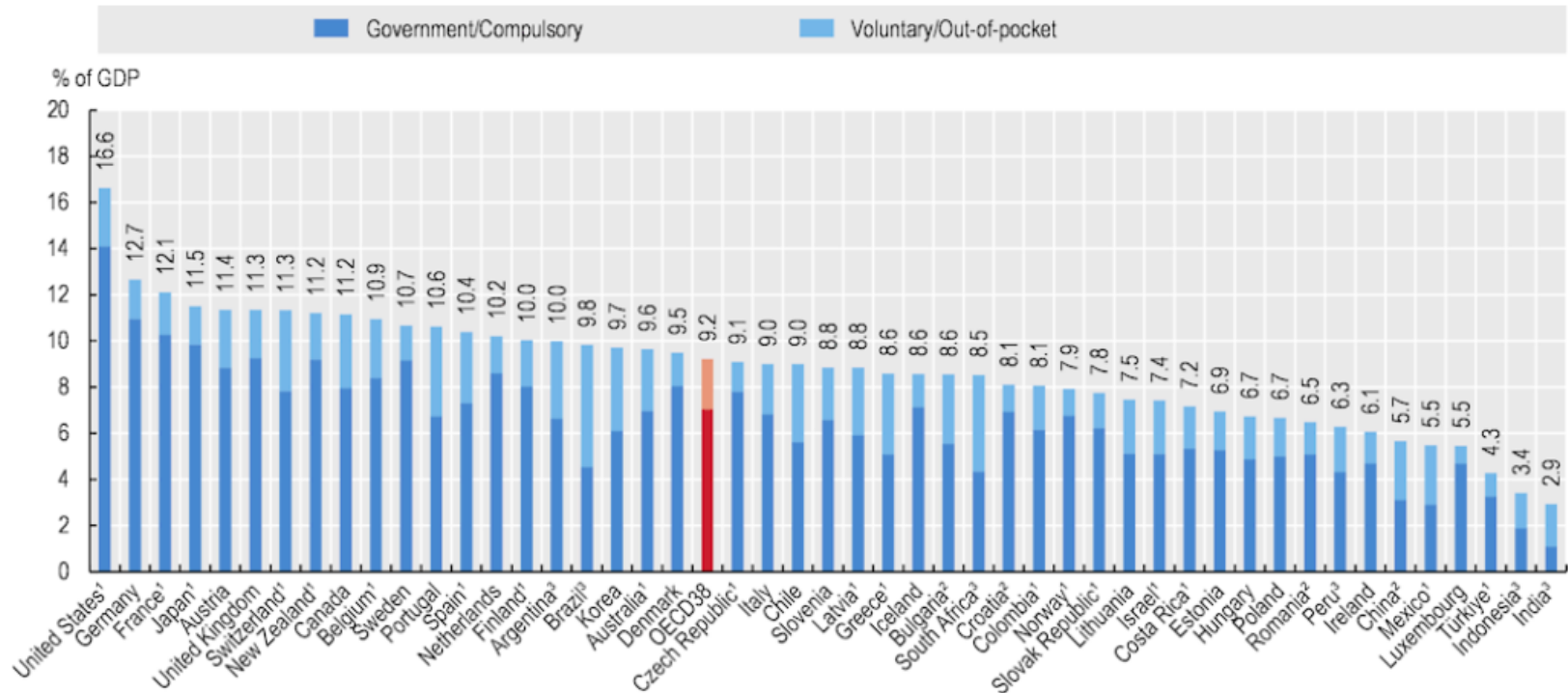
BIEM
Spring 2026



Healthcare System
Gruber - Chapters 15&16

Health Care Spending in the OECD Nations, 2022

Health expenditure as a share of GDP, 2022 (or nearest year)



Source: OECD Health Statistics 2023; WHO Global Health Expenditure Database

Different models around the World

- **National health insurance:** funded by general taxation, provided by the government, universal access. Examples: UK, Italy.
- **Social health insurance:** funded by payroll contributions (employers + employees), provided mostly by private providers, universal access through insurance funds. The government regulates prices.
- **Single-payer:** health insurance funded by taxes and provided by the government.
- **Private insurance model:** funded by private insurance and provided by private sector. Example: US.

How Health Insurance Works: The Basics

- Individuals, or firms on their behalf, pay monthly premiums to insurance companies.
- In return, the insurance companies pay the providers of medical goods and services for most of the cost of goods and services used by the individual.
- There are three types of patient payments:
 - Deductibles—limit to cost individual pays
 - Copayment—fixed payment individual pays
 - Coinsurance—percentage of each bill individual pays

Distribution of the U.S. Population Across Health Insurance Types

	People (millions)
Total population	324.6
<i>Private</i>	220.8
Employment-based	183.0
Direct purchase	33.2
<i>Public</i>	110.7
Medicare	58.8
Medicaid	55.9
Tricare/CHAMPVA	3.2
<i>The uninsured</i>	26.1
Data from: Keisler-Starkey and Bunch (2020), Table 1.	

Private Insurance

- In 2019, about 68.5% of the U.S. population had private health insurance.
- Private insurance is provided by employers and by the nongroup insurance market.
- **Nongroup insurance market:** The market through which individuals or families buy insurance directly rather than through a group, such as the workplace.

Why do Employers Provide Private Insurance?

Motive I: Risk Pooling

- One reason employers provide insurance is to pool risks.
 - Risk pool: The group of individuals who enroll in an insurance plan.
- The goal of all insurers is to create *large insurance pools with a predictable distribution of medical risk*.
- The *law of large numbers* helps achieve this goal.
- By pooling all employees, employer-provided health insurance also avoids adverse selection.

Why do Employers Provide Private Insurance?

Motive II: The Tax Subsidy

- Employers also provide insurance because it is subsidized.
- **Tax subsidy to employer-provided health insurance:** Workers are taxed on their wage compensation but not on compensation in the form of health insurance, leading to a subsidy to health insurance provided through employers.
- Example:

	Marginal Product, Wage	Employer Health Insurance Spending	Pre-Tax Wage	After-Tax Wage	Personal Health Insurance Spending	After-Tax, After-Health Insurance Income
Nigel	\$30,000	0	\$30,000	\$20,000	\$4,000	\$16,000
Kim	30,000	\$5,000	25,000	16,666	0	16,666

- While Nigel's private insurance is cheaper, Khadija ends up with more income after taxes due to the subsidy to employer-provided insurance.

Why do Employers Provide Private Insurance?

Motive II: The Tax Subsidy

The subsidy to employer-provided health insurance is generally not well understood.

- Subsidy to employees, not employers.
- Employer is indifferent between payments in wages and in health insurance (both are tax deductible).
- Worker prefers to be paid in health insurance rather than wages since this reduces the worker's tax payments.
- To end the tax subsidy, don't increase the corporate tax paid by the firm; instead, include employer spending on health insurance as part of an employee's taxable income.

The Other Alternative: Nongroup Insurance (individual)

- The nongroup insurance market was traditionally not a well-functioning market.
- Nongroup insurance was not always available.
- **Those in the worst health were often unable to obtain coverage** (or obtain it only at an incredibly high price).

Medicare

- Medicare: A federal program that provides health insurance to **all people over age 65** and **disabled persons under age 65**.

Every citizen who has worked for 10 years in Medicare-covered employment (and their spouse) is eligible for Medicare at age 65.

Medicare is financed by a payroll tax on employees and employers.

Medicaid

- **Medicaid:** A federal and state program that provides healthcare for the **poor**.
- Medicaid benefits are targeted at several groups:
 - Those who qualify for cash welfare programs
 - Most low-income children in the United States
 - Most low-income pregnant women
 - All very low-income families (in states that expanded the program to this group under ACA.)
 - The low-income elderly and disabled (for expenses not covered by Medicare).

TRICARE/CHAMPVA

- **TRICARE**

- Administered by the Department of Defense
- Serves military retirees and the families of active-duty, retired, or deceased service members

- **CHAMPVA**

- Civilian Health and Medical Program for the Department of Veterans Affairs
- A health care program for disabled dependents of veterans and certain survivors of veterans

The Uninsured

→ Who are they?

- There are 26 million in the United States without any insurance coverage.
- The uninsured have low-than-average incomes.
- In 2019, nearly three quarters of the nonelderly uninsured came from families where one or more members were full-time workers.
- About 14.2% of the uninsured are children.

→ Why are some individuals uninsured?

- They may be counting on uncompensated care.
 - Uncompensated care: The costs of delivering health care for which providers are not reimbursed.
- Insurance may cost too much, given risks and prices.
- Insurers may be unwilling to insure the worst risks because of fears of adverse selection.
- They are not appropriately valuing insurance coverage.

Why Care About the Uninsured?

1. There are physical externalities associated with communicable diseases.
2. There is a significant financial externality imposed by the uninsured on the insured.
3. Care is not delivered appropriately to the uninsured.
4. Paternalism and equity motivations.
5. A final reason for caring about the uninsured is that becoming uninsured is a concern for millions of individuals who currently have insurance → **Job lock**

Job lock: The unwillingness to move to a better job for fear of losing health insurance.

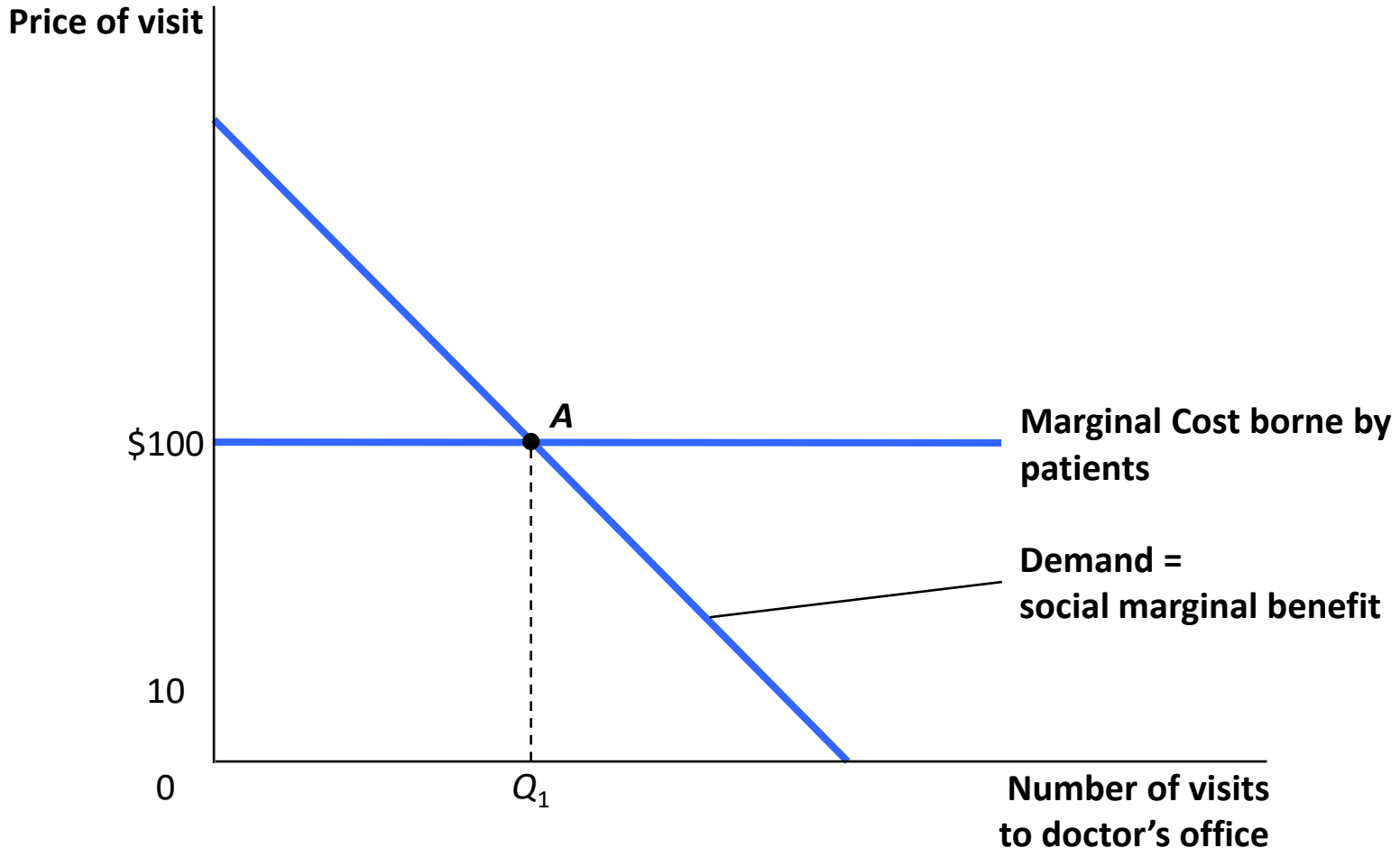
- Health insurance availability may inhibit productivity-increasing job switches.
- Overall, research shows that job lock appears **quantitatively important**.

How Generous Should Insurance Be to Patients?

- Most generous plans (to patients) provide first-dollar coverage.
 - First-dollar coverage: Insurance plans that cover all medical spending, with little or no patient payment.
- The consumption-smoothing benefit from first-dollar coverage of minor and predictable medical events is small for two reasons:
 - Risk-averse individuals gain little utility from insuring a small risk.
 - Individuals are much more able to self-insure such spending than to self-insure large and unpredictable medical events.
- On the other hand, the **moral hazard costs** are large.

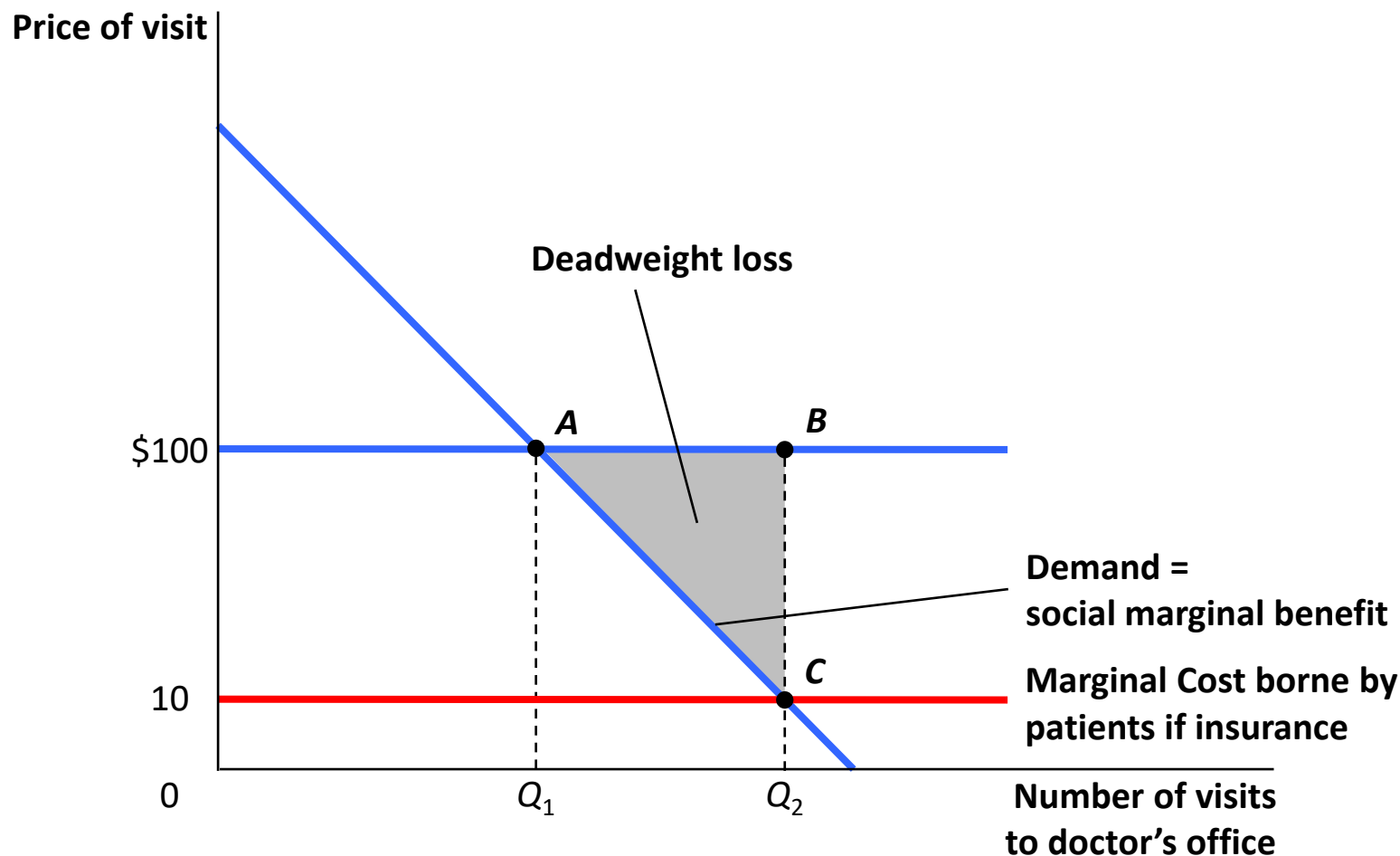
Moral Hazard Costs of Health Insurance for Patients

Trade-off of health insurance: the gains in terms of consumption smoothing versus the costs in terms of overuse of medical care



Moral Hazard Costs of Health Insurance for Patients

Trade-off of health insurance: the gains in terms of consumption smoothing versus the costs in terms of overuse of medical care



Optimal Health Insurance

Optimal health insurance:

- Trades off moral hazard against risk protection.
- First-dollar coverage bad for moral hazard, not very valuable risk protection.
- Therefore, optimal health insurance policy:
 - Individuals bear a large share of medical costs within some affordable range.
 - Fully insured only against very large costs.
- Most coverage appears more generous than is optimal, and many people don't have coverage.

→ Why are people either uninsured or “overinsured”?

How Generous Should Insurance Be to Medical Providers?

- Insurance often determines how much medical providers are paid.
- One payment system is retrospective reimbursement.
 - **Retrospective reimbursement: Reimbursing physicians for the costs they have already incurred.**
- Encourages overutilization since providers are paid regardless of necessity or value of care.

Managed Care

- **Managed care: An approach to controlling medical costs using supply-side restrictions, such as limited choice of medical provider.**
- One form of managed care are **preferred provider organizations**.
 - Preferred provider organization (PPO): A healthcare organization that lowers care costs by shopping for health care providers on behalf of the insured.
- **Health maintenance organizations** are a second kind of managed care organization.
 - Health maintenance organization (HMO): A healthcare organization that integrates insurance and delivery of care by, for example, paying its own doctors and hospitals a salary independent of the amount of care they deliver.
 - In the classic staff model, HMOs hire their own physicians and may have their own hospitals.

Managed Care and Prospective Reimbursement

- Managed care organizations are paid by **prospective** reimbursement, not retrospective.
- **Prospective reimbursement: The practice of paying providers based on what treating patients should cost, not on what the provider spends.**
- Payment is the same regardless of spending, so there is an incentive to reduce costs.
- Just as retrospective reimbursement offers incentives for excessive care, **prospective reimbursement offers incentives for insufficient care.**
 - By delivering less care, the physician pockets a larger share of the payment.

Health Care Reform and the ACA

There has been an historical impasse over national health insurance.

- Some argue for a single-payer system.
 - Government-provided health insurance for all.
 - Guarantees full coverage.
 - Low administrative costs.
 - Eliminates inequality in care.
 - Straightforward to control costs by budgeting.

Health Care Reform and the ACA

- Public system also has disadvantages:
 - Dramatically increases government expenditures.
 - Budgeting is a blunt instrument.
 - May not allow doctors to use a technology that is worth its high cost.
 - Severe political hurdles from health insurance companies.
- Others push for a private-sector solution, possibly with subsidies.
 - Adverse selection, other market failures remain.
 - No evidence that the private sector can contain costs.

The Massachusetts Experiment

- In 2006, Massachusetts pushed to cover remaining 9% without insurance.
- “Three-legged-stool” approach:
 - 1) Ban pre-existing conditions exclusion, health-based pricing.
 - 2) Individual mandate, avoiding adverse selection.
 - Mandate: A legal requirement for employers to offer insurance or for individuals to obtain some type of insurance coverage.
 - 3) Subsidized/free insurance for low-income families.
- Results:
 - Massachusetts uninsurance rate 3%, compared to 18% nationally.
 - Half of the increase in coverage from Medicaid or government subsidized plans.
 - Premiums in the nongroup market have fallen by half relative to national trends.
 - Costs of the reform roughly consistent with projections.
 - Some studies have found the policy has improved health of population.

The Affordable Care Act

- In 2010, President Obama signed into law the Affordable Care Act, based on the Massachusetts health reform.
 - Adopts the “three-legged stool” of Massachusetts:
 - Prices are community rated, not health-specific.
 - Individuals are mandated to buy insurance.
 - Medicaid expanded, with subsidies for lower-income people:
- There was a massive expansion of the Medicaid program put in place by the ACA, which extended eligibility to all families, regardless of family structure, below 138% of the federal poverty line. (As of 2020, 12 states have opted out of this expansion)

APPLICATION: Rising Health Care Costs and Cost Control Efforts in the ACA

- Since 1950, the price index for medical care has risen by 2.1 percentage points more per year than inflation, but much of the rapid rise in spending has been quality-improving technological change.
- Limiting spending to efficient uses would reduce costs but is hard to do. The ACA has five approaches.
 1. Cap tax break of employer-provided insurance.
 2. Increase competition for health insurance.
 3. “Depoliticize” rate setting for providers.
 4. Sponsor “comparative effectiveness” research.
 5. Promote pilot programs of alternative structure for organizing and reimbursing providers.

Early Evidence on the Effects of the ACA

Projecting the impacts of a reform as large as the ACA is difficult, but the CBO attempted to do so, most recently in CBO (2014).

- They projected that these reforms would lead to 26 million newly insured residents by 2019.
- The CBO also projected that the spending cuts and revenue increases in the ACA would more than offset the new spending under the ACA so that the law will reduce the deficit by more than \$100 billion over the first decade and more than \$1 trillion over the next.

Percentage Uninsured in the U.S., by Quarter

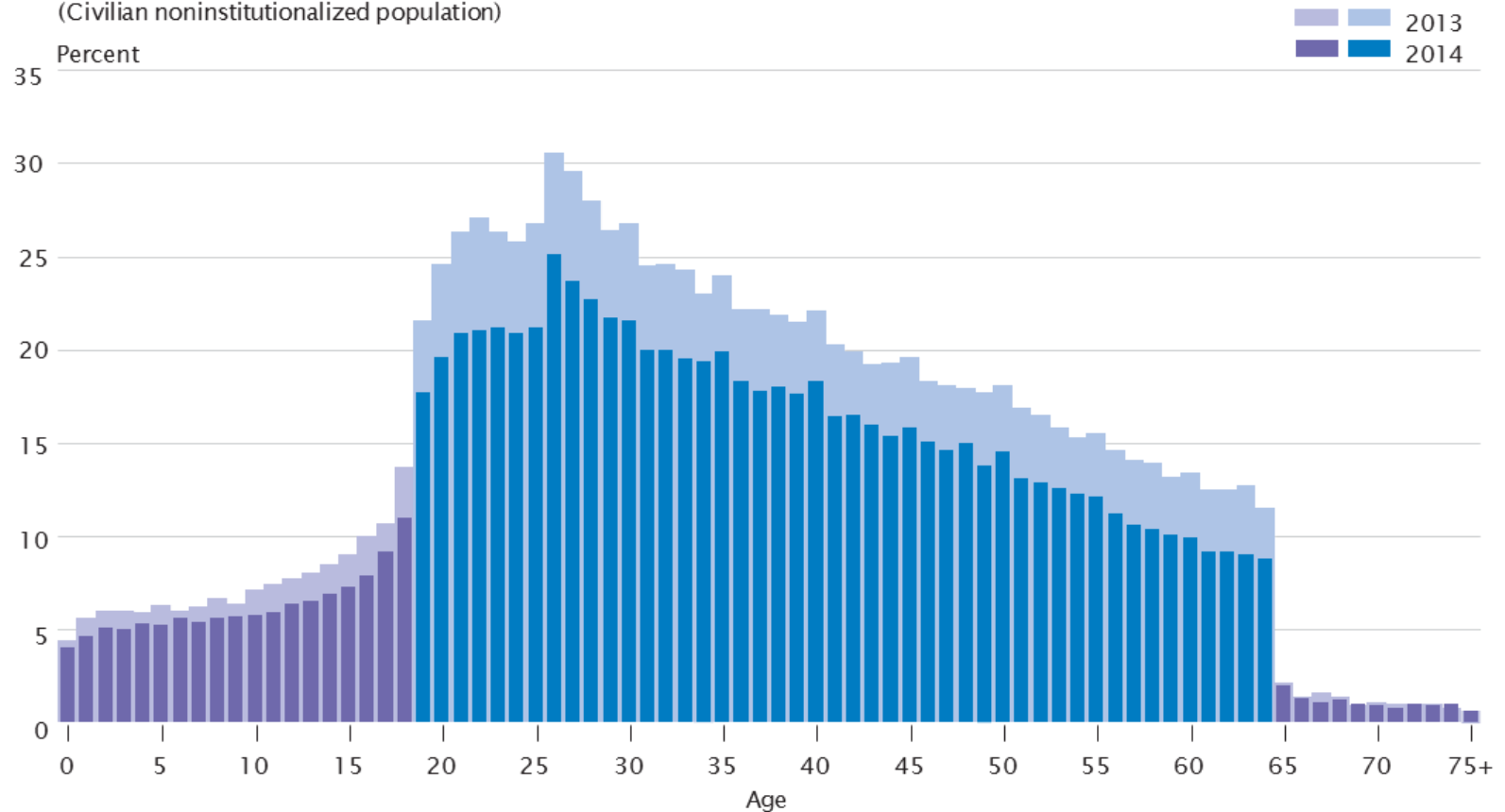
Do you have health insurance coverage? Among adults aged 18 and older

■ % Uninsured



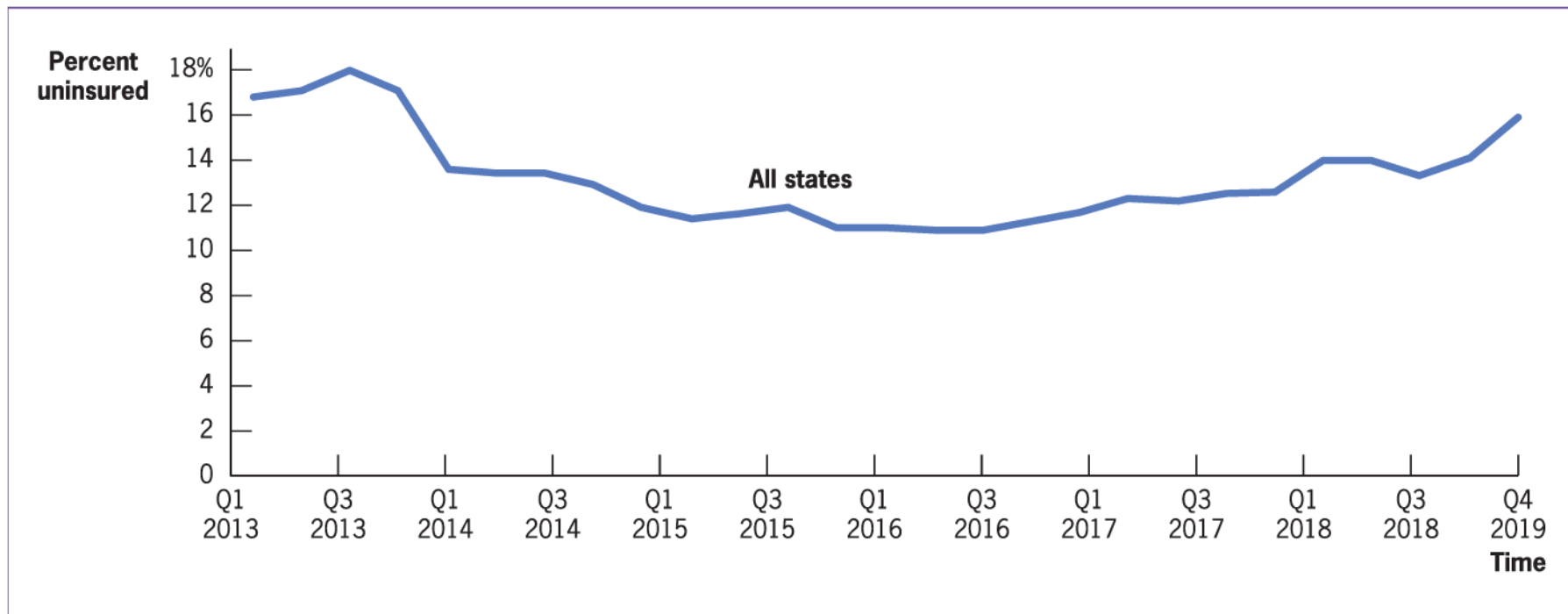
SOURCE: GALLUP-HEALTHWAYS WELL-BEING INDEX

Figure 4.
Uninsured Rate by Single Year of Age: 2013 and 2014
(Civilian noninstitutionalized population)



For information on confidentiality protection, sampling error, nonsampling error, and definitions in the American Community Survey, see <www2.census.gov/programs-surveys/acs/tech_docs/accuracy/ACS_Accuracy_of_Data_2014.pdf>.

Source: U.S. Census Bureau, 2013 and 2014 1-Year American Community Surveys.



Uninsurance rates among the states declined precipitously from 2014 through 2016 before rising again in 2017.

Projected Impacts of the ACA and Early Evidence on Its Effects

Early evidence on the effects of the ACA appears to support the contentions of the CBO analysis.

- The number of uninsured in the United States has fallen by about 20 million, with the uninsurance rate declining by more than one-third.
- Other studies suggest that this coverage has also improved access to care and self-reported health, decreased emergency room use, and increased use of preventive care.
- Cost growth since the passage of the ACA has been historically low, at a rate of 1.4% in 2013.

The ACA Runs Into Trouble

- Public support for the ACA was below 50% both before and after passage and remained below 50% even as the law was enacted and the coverage gains took place.
 - This unpopularity was a significant factor in major Republican gains in the 2016 election.
- The ACA's partial reform structure created more opponents than supporters.
 - Many of the benefits were through expanded Medicaid and other mechanisms not directly linked by voters to "Obamacare."
- The Trump administration and Congress took actions that significantly weakened the ACA.
 - Foremost of these actions is the repeal of the individual mandate.

The Italian health-care system

- The SSN (**Sistema Sanitario Nazionale**) was introduced in 1978. Before 1978: insurance- based system.
- SSN: **Universal coverage**; funded through general taxation.
- According to international rankings it is one of the best health-care system in the world (health expenditure/GDP vs results).
- The SSN is a regionally-based national health service. It provides free of charge universal coverage at the point of service.
- While the **national level** ensures the **general objectives** and **fundamental principles** of the national health care system are met, **regional governments** are responsible for ensuring the **delivery of a benefits package** to the population
- After the 1992 Reform → introduction of «**quasi-mkt**» features:
 - Separation between the health-care purchase and provision
 - Competition between public providers and private providers (which must satisfy quality standards defined by the State)

4 levels:

1) Central government (specifically, the Health Ministry):

- controls the distribution of tax revenue for publicly financed health care and defines a national statutory benefits package to be offered to all residents in every region, the “essential levels of care” (*livelli essenziali di assistenza*, LEA)
- is responsible for the general guidelines for prevention, diagnosis and therapies
- coordinates and monitors Regions’ health activities.

2) Regions:

- enjoy significant autonomy in determining the macro structure of their health systems
- organize and deliver health services through local health units (ASL)

3) ASL (*Aziende Sanitarie Locali*):

- are managed by a general manager appointed by the governor of the region
- are responsible for territorial planning
- purchase and deliver primary care, hospital care, outpatient specialist care and public health care
- Funded through a fixed reimbursement per-patient

4) Health-care providers (hospitals and physicians):

- Public hospitals either are managed directly by the local health units or operate as semi-independent public enterprises.
- Hospitals funded through «perspective payments» based on DRGs: Diagnosis Related Group. Each hospital admission is classified into a group based on the patient's diagnosis and treatment, and the hospital receives a fixed, pre-set payment for that group, regardless of the actual cost of care.
- General practitioners: they are paid a fixed amount for each patient (max 1500 patients)
- Hospital doctors/practitioners: salary determined by the State; varies with age and specialization

Exercises

1) Which of the following statements correctly describes the difference between prospective and retrospective reimbursement of health care providers?

- a) Retrospective reimbursement pays providers a fixed amount per patient regardless of costs
- b) Prospective reimbursement reimburses providers based on actual costs incurred
- c) Retrospective reimbursement creates incentives for overutilization of medical care
- d) Prospective reimbursement fully eliminates incentives for inefficient care provision

2) Which statement correctly describes the tax subsidy to employer-provided health insurance?

- a) It subsidizes employers directly through corporate tax reductions
- b) It encourages firms to offer insurance instead of wages
- c) It benefits workers by excluding insurance from taxable income
- d) It reduces government revenue only through payroll taxes

3) Which policy tool in the Affordable Care Act was primarily aimed at reducing adverse selection in insurance markets?

- a) Expansion of Medicaid
- b) Community rating
- c) Individual mandate
- d) Comparative effectiveness research

4) Doctor's checkup have a marginal cost of 150 each. Aggregate demand on part of consumers is represented by the following function: $Q = 200 - P$. What unitary subsidy should the government put in place if the target is achieving $Q = 100$ visits?

- a) 50\$
- b) 100\$
- c) 150\$
- d) 25\$

5) Which of the following best explains why health insurance can lead to moral hazard?

- a) Insured individuals face higher prices for medical care
- b) Insurance reduces uncertainty about future income
- c) Insured individuals do not bear the full marginal cost of care
- d) Insurance eliminates adverse selection

T/F questions (please provide an explanation)

- 1) If the government sets a subsidy equal to marginal cost, the quantity of medical visits demanded will always be socially optimal.
- 2) Subsidizing doctor visits can generate moral hazard by encouraging excessive use of medical care.
- 3) The Italian National Health Service (SSN) guarantees universal coverage that is free at the point of service and financed mainly through general taxation.

EXERCISE 1

The demand function for antibiotics is as follows: $q=30-p$. The marginal cost of producing each antibiotic is 10\$.

- a) Please compute the market equilibrium.
- b) If the state subsidizes antibiotics by paying half of each of them (5\$) and therefore allowing them to be purchased at a discounted price, what would be the new equilibrium quantity ? Is there going to be any deadweight welfare loss ? If so, please compute it.

EXERCISE 2

Briefly explain the main trade-off policymakers face when designing health insurance coverage. In your answer, refer to risk protection and moral hazard.